



OpenMatter®

Quantum-Resistant Secrets for Trustless Decentralized Applications

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Motivation and Problem

Setting: Decentralized compute marketplace (OpenMatter Network)

- Users deploy workload configurations on-chain with encrypted environment variables
- Providers execute workloads and need plaintext secrets at runtime

Problem: How to deliver secrets to providers without :

1. A single trusted party that can decrypt everything
2. Exposing secrets on-chain at any point
3. Requiring provider identity at encryption time

Solution: Threshold decryption by a decentralized Key Generation Committee (KGC), built entirely on post-quantum lattice assumptions.



Requirements

R1. On-chain confidentiality — ciphertext reveals nothing to any observer.

R2. Creation-time encryption — secrets encrypted at deployment submission, before provider assignment.

R3. Provider ignorance — assigned provider learns nothing until explicitly granted access.

R4. Per-deployment isolation — compromise of one deployment reveals nothing about any other, even to the same provider.

R5. No single point of trust — no single entity can decrypt all secrets.

R6. Forward secrecy — after zeroization, secrets are irrecoverable



OPENMATTER · PROTOCOL ROLES

Who holds what, and who does what

PARTY

ROLE



User

Encrypts secrets



Provider

Runs the workload



Chain

Public ledger



Committee

Threshold decryption (any t of n)

SETUP

once, rarely

Committee

Phase 0 — the committee jointly creates the key and publishes the shared public key.

Phase 1 — the user encrypts secrets and publishes them to the chain.

DEPLOY

every job

User

1 · encrypted secret

Chain

2 · assign provider

Provider

4 · combine + decrypt

Committee

3 · partial decryptions

Key material

- Committee — each node holds a piece of the private key
- User — a signing keypair (signs every request)
- Provider — gets the decrypted secret, only for the run



OPENMATTER · CRYPTOGRAPHIC FOUNDATION

Built on Ring-LWE

DEFINITION

Let $R = \mathbb{Z}[X] / \Phi_m(X)$ be a cyclotomic ring of degree $\phi(m)$, modulus q , $R_q = R / qR$.

For a secret $s \leftarrow \chi$, a uniform $a \in R_q$, and small error $e \leftarrow \chi$:

$(a, b = a \cdot s + e) \approx_c$ indistinguishable from uniform over $R_q \times R_q$.

INSTANTIATION

- Ring degree $\phi(m) = 4098$ (cyclotomic index $m = 4099$)
- Modulus q : 109-bit prime, $q \equiv 1 \pmod{m}$
- Plaintext modulus $t = 2$
- Secret $CB(1)$, error $CB(20)$ (centered binomial)
- Security ≈ 128 -bit classical / ≈ 120 -bit post-quantum

KEY GENERATION

$sk = s \leftarrow CB(1)$

$pk = (a, b = -a \cdot s + e)$

$a \in R_q$ uniform, $e \leftarrow CB(20)$

No trapdoor — security rests on a hard lattice problem.

Post-quantum hardness, by construction.



OPENMATTER · KEY GENERATION COMMITTEE

Decryption by committee

Committee nodes — a quorum decrypts together



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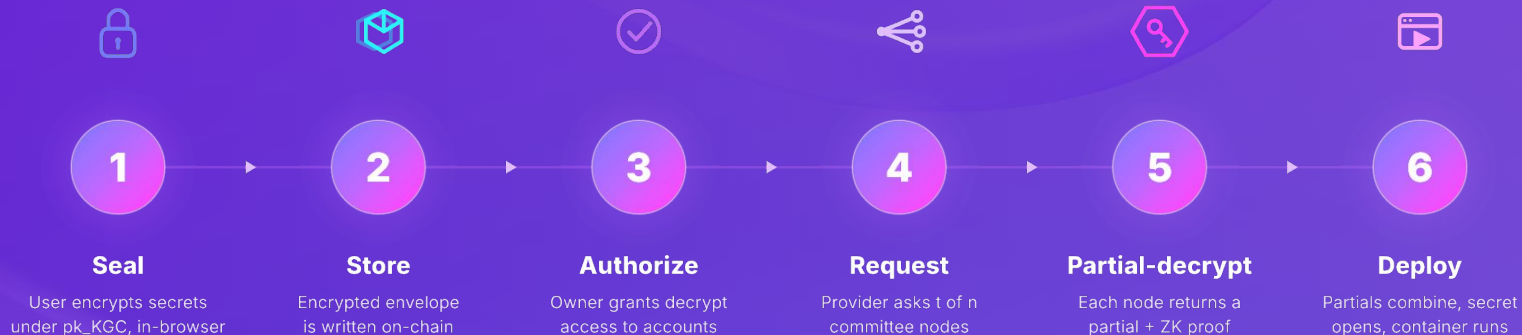
No single node ever holds the full private key.

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HOW IT WORKS

Secrets that decrypt only when authorized

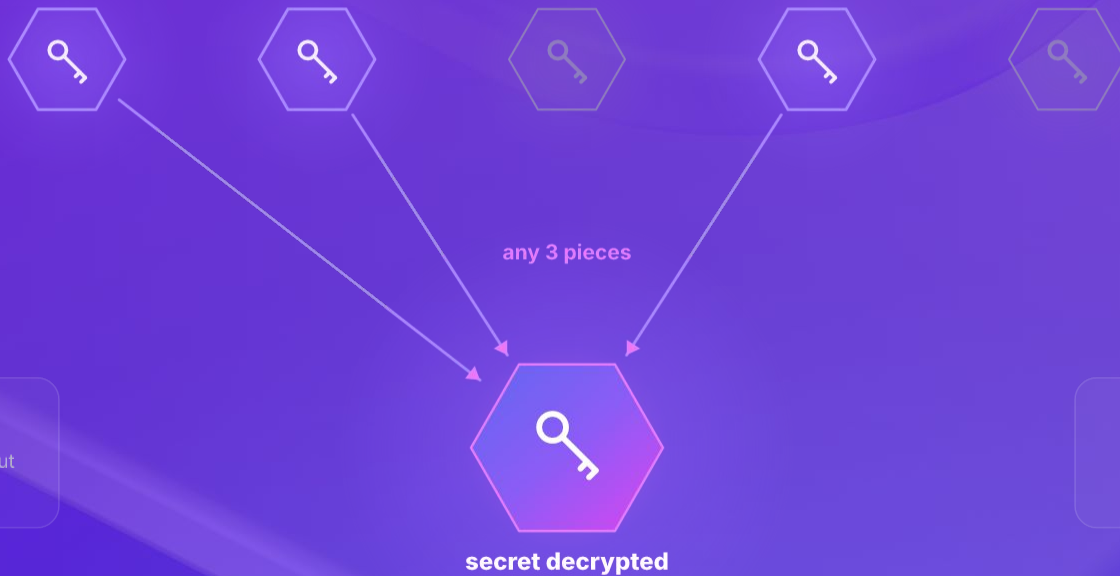




THE QUORUM GUARANTEE

Any 3 of 5. Never one.

5 committee nodes — each holds one piece of the private key



Any 2 pieces

reveal nothing about
the secret



No single node

ever holds the full
private key



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Live Q/A
Thank you.

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